Macroeconomic Data Analysis

ECON 112 - Spring 2026

University of California, San Diego

Teaching Team

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Syllabus

Note: This syllabus covers all the important organizational details. I consider the Syllabus my contract with you and I commit to uphold it as written. While I do not anticipate doing so, situations may arise that require adjustments of the class rules to ensure that the quality of your learning is preserved, so please consider the information contained in the Syllabus as subject to revision.

The courses examines time series methods for data analysis with an emphasis on macroeconomic applications. The aim of the class is to provide students with a basic formal introduction to modern time series techniques, hands-on experience in applying them to real-world macroeconomic data, and understanding of how to bring data to bear on policy-relevant macroeconomic theories.

Logistics and Important Dates

We meet twice a week on Tuesdays and Thursdays from 12:30 pm to 1:50 pm, Pacific Time in Coalition 122. Eugenio, our TA, will schedule review sessions ahead of exams. More information about review sessions and office hours will be provided in Week 1.

There will be three in-class quizzes and one final exam. Please mark your calendar as follows:

- **Quiz 1, Thu Apr 16th, class time;**
- **Quiz 2, Tue May 5th, class time;**
- **Quiz 3, Tue May 26th, class time;**
- **Final Exam, Monday June 8th, 11:30 am**

Textbook

The textbook for the class is "*Applied Time Series for Macroeconomics*," by Hilde Bjornland and Leif Thorsrud, Gyldendal Publisher (2015).

Statistical Software

For our statistical computations we will be using R. R is a programming language and a free and open-source software environment that is widely used worldwide. The basic software can be downloaded at https://repo.miserver.it.umich.edu/cran/. I recommend operating R using RStudio Desktop, a free and open-source integrated development environment for R, downloadable at https://www.rstudio.com/.

Homework Assignments and Practice Problems

I will assign 5 Homework Assignments that will be graded. The lowest score of the 5 will be dropped. The purpose of Homework Assignments is twofold: (i) practice on the theoretical concepts covered in class; (ii) get you working on real data with the tools that are presented in class. The dates when the Homework Assignments will be posted and due are listed in the Class Schedule at the end of this Syllabus.

I will also post 4 Practice Problem Sets, one before each of the three quizzes and one before the final exam. The practice problem sets will not be graded, and solutions will be provided ahead of the tests. The purpose of the Practice Problem Sets is to have you practice on the material covered in class on which you will be tested. You will have the chance to review the solutions of the Practice Problem Sets during review sessions held by our TA.

Writing Assignments and Recorded Presentation

There will be two "Writing Assignments", one due in Week 5 and one due in Finals Week. These assignments contain both a numerical component and a writing component. The purpose of the assignments is to help you develop the skills necessary to turn numerical results into well-articulated written explanations, and to showcase your critical thinking about using the tools you learned to interpret your results.

In addition, together with the second writing assignment, you will be asked to submit a short video of a presentation of the results contained in your writing assignment. More detailed on the presentation will be provided in class.

Discussion Board

We will be using Piazza for class discussion. Rather than emailing questions to me or Eugenio, we encourage you to post your questions on Piazza. You can find our Piazza page on Canvas or here

Grading

Your final grade will be determined according to the following percentages

Participation	5%	
Quizzes	30%	(10% each)
Final Exam	10%	
Homework Assignments	20%	(5% each, lowest of 5 dropped)
Writing Assignments	35%	(WA 1 10%, WA 2 25%)

Participation. Participation points will be based on submitting the answer to three reflection questions asked at the end of each week. Submissions are due every Monday by noon, Pacific Time. Of the total 10 submissions, only 8 will count towards participation points (i.e. you can skip two submissions if you'd like with no consequences on your participation points).

Exam Modality All exams to be held in-person. If changes to this plan are needed, they will be promptly communicated.

Academic Integrity

I strongly value creating a fair and safe community among students in my classes. I am committed to uphold academic integrity as a fundamental principle in our interactions. *You, as student, are expected to do your own work.* Cheating, plagiarism, fabrication, lying, bribery, threatening behavior and assistance to acts of academic dishonesty are examples of behaviors that violate academic integrity. All students of UC San Diego are responsible for knowing and adhering to the UCSD policy on Academic Integrity. Ordinarily, a student engaged in any act of academic dishonest will receive a failing grade for the course. In addition, all incidents of academic misconduct shall be reported to the Academic Integrity Office. Depending on its findings, students who are found to be in violation of the academic integrity policy will be subject to non-academic sanctions, including but not limited to university probation, suspension, or expulsion. The Academic Integrity Office can be contacted by email at aio@ucsd.edu or by telephone at 858-822-2163. Additional information regarding the University Academic Integrity policy is available at https://academicintegrity.ucsd.edu/.

Missed Examinations

No make-up exams will be given in this class. You must take the final exam in order to receive a grade in this course. The date is written above, be sure to mark your calendar. If you miss a quiz without a university accepted excuse, you will receive a grade of zero (0) on that test. If you miss a quiz with a university accepted excuse, the weight of the final exam will be increased accordingly. Excuses for missed exams must be pre-approved by professor Rondina (except when this is not possible in an emergency situation). A student who misses an exam due to physical illness will be required, when possible, to provide documentation from a health care professional indicating why the student was

physically unable to take the exam. All documentation must be provided to professor Rondina as soon as reasonably possible. Failure to comply with any of the above in the specified manner will result in a grade of zero (0) for the exam.

University Policies and Resources

Conduct Code. To foster the best possible working and learning environment, UC San Diego strives to maintain a climate of fairness, cooperation, and professionalism. UC San Diego's Principles of Community illustrate the expectations of all members of our community. I recommend reviewing the the Student Conduct Policies and Procedurs to make sure you are familiar with the behavior that is expected from you in class and on campus in general.

Disability Accommodations. Campus policy regarding disabilities requires that faculty adhere to the recommendations of the Office for Students with Disabilities (OSD). Any student eligible for and needing academic adjustments or accommodations because of disability should submit to me a letter from OSD describing appropriate adjustments or accommodations and should arrange to meet with me as soon as possible so that arrangements can be made in a timely manner. University policies regarding disabilities are available at https://osd.ucsd.edu/. Appointments with OSD (phone or in-person) can be made by calling 858.534.4382 or by email at osd@ucsd.edu.

Religious Observance. Campus policy regarding religious observances requires that faculty make every effort to reasonably and fairly accommodate all students who, because of religious obligations, have conflicts with scheduled examinations, assignments or required attendance. See full details of policies on examinations here. If you have scheduling conflict covered by this policy, please let me know as soon as possible so that we can reschedule the relevant assignment/examination.

Harassment Policy. The University Policy on Discrimination and Harassment applies to all students, staff and faculty. Any student, staff member orfaculty member who believes (s)he has been the subject of discrimination or harassment based on race, color, national origin, sex, pregnancy, age, disability, creed, religion, sexual orientation, gender identity, gender expression, veteran status, political affiliation, or political philosophy, should contact the Office for the Prevention of Harassment and Discrimination (OPHD) at (858) 534-8298, ophd@ucsd.edu, or reportbias.ucsd.edu.

Data Privacy. The University adheres to the standards for student privacy rights and requirements as stipulated in the Federal Rights and Privacy Act (FERPA) of 1974, see http://ucsd.edu/catalog/front/ferpa.html.

Counseling. Managing the many challenges of being a college student can be very stressful. Always remember that talking to somebody that is professionally trained to help is just an email, or short walk, away: https://caps.ucsd.edu/.

Econ 112 - Tentative Class Schedule - Spring 2026

Week	**Date**	**Topic**	**Assignments**
Week 1	Lec. 1 (Mar 31)	Intro + R + Math Review	
	Lec. 2 (Apr 2)	OLS, Hypothesis Testing	HW 1 Post
Week 2	Lec. 3 (Apr 7)	MA Models	
	Lec. 4 (Apr 9)	MA and AR Models	HW 1 Due, PPS 1 Post
Week 3	Lec. 5 (Apr 14)	AR Models	HW 2 Post
	Lec. 6 (Apr 16)	**Quiz 1** + Estimating AR Models	**WA 1** Post
Week 4	Lec. 7 (Apr 21)	Forecasting	HW 2 Due
	Lec. 8 (Apr 23)	Nonstationarity	
Week 5	Lec. 9 (Apr 28)	Trends and Cycles	**WA 1** Due, PPS 2 Post
	Lec. 10 (Apr 30)	Trends and Cycles	
Week 6	Lec. 11 (May 5)	**Quiz 2** + Trends and Cycles	HW 3 Post
	Lec. 12 (May 7)	Endogeneity Bias	
Week 7	Lec. 13 (May 12)	Vector Autoregression (VAR)	HW 3 Due, HW 4 Post
	Lec. 14 (May 14)	Vector Autoregression (VAR)	
Week 8	Lec. 15 (May 19)	Structural VAR	HW 4 Due, PPS 3 Post
	Lec. 16 (May 21)	Structural VAR	
Week 9	Lec. 17 (May 26)	**Quiz 3** + Structural VAR	**WA 2** Post
	Lec. 18 (May 28)	Structural VAR	
Week 10	Lec. 19 (Jun 2)	Spurious Regressions	HW 5 Post
	Lec. 20 (Jun 4)	Cointegration	HW 5 Due, PPS 4 Post
Final Exam	**Mon Jun 8**	**11:30 am, place TBD**	
WA 2 + Pres	Due on Friday **Jun 12**		

HW = Homework Assignment

WA = Writing Assignment

PPS = Practice Problem Set